

FACULTY OF ENGINEERING
DIPLOMA ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY



Introduction to Machine Learning (BE103711)

QUESTION BANK

SECTION A: 3 MARKS QUESTIONS (25 Questions)

UNIT 1: Introduction to Machine Learning:

1. Define **Machine Learning**. Explain how it differs from Human Learning.
2. What are the **types of Machine Learning**? Name and briefly explain them.
3. List any **three applications of Machine Learning** with examples.
4. Explain the role of **Machine Learning tools and technologies** in developing ML models.
5. Differentiate between **Human Learning** and **Machine Learning**.

UNIT 2: Preparing to Model:

1. Explain the **Machine Learning activities** involved in building an ML model.
2. What are the **different types of data** used in Machine Learning?
3. Explain the **structure of data** in Machine Learning with examples.
4. What is **Data Preprocessing**? Why is it important in Machine Learning?
5. Define **Dimensionality Reduction** and **Feature Subset Selection**.

UNIT 3: Modeling and Evaluation:

1. Differentiate between **Predictive Models** and **Descriptive Models**.
2. Explain the **process of training a supervised learning model**.
3. What is **Model Representation**? Why is it important in Machine Learning?
4. What is **Model Interpretability**? Mention its significance.
5. List any **three methods to improve the performance of a Machine Learning model**.

UNIT 4: Basics of Feature Engineering:

1. Define **Feature** and **Feature Engineering**. Explain their importance in Machine Learning.
2. What is **Feature Transformation**? Mention its types.
3. Differentiate between **Feature Construction** and **Feature Extraction**.
4. What are the **issues in high-dimensional data**?
5. What is **Feature Subset Selection**? State its advantages.

UNIT 5: Overview of Probability:

1. Explain the importance of **Statistical Tools** in Machine Learning.
2. Define **Random Variable**. Differentiate between **Discrete** and **Continuous Random Variables**.
3. What is the **Central Limit Theorem (CLT)**? State its significance.
4. What is **Hypothesis Testing**? Mention its basic steps.
5. Define **Monte Carlo Approximation** and mention one of its applications.

UNIT 6: Bayesian Concept Learning:

1. What is the **importance of Bayesian methods** in Machine Learning?
2. State **Bayes' Theorem** and explain its components.
3. What is **Concept Learning**? Explain its relationship with **Bayes' Theorem**.
4. Define a **Bayesian Belief Network (BBN)**.
5. List any **three applications of Bayesian methods** in Machine Learning.

UNIT 7: Supervised Learning:

1. Define **Supervised Learning**. Mention its main characteristics.
2. Differentiate between **Classification** and **Regression**.
3. Explain the **steps involved in supervised learning**.
4. What is a **Classification Model**? Give two examples.
5. Name any **four Classification and Regression algorithms** used in Machine Learning.

UNIT 8: Unsupervised Learning:

1. Define **Unsupervised Learning**. Mention its main characteristics.
2. Differentiate between **Supervised Learning** and **Unsupervised Learning**.
3. What is **Clustering**? Give two real-world applications.
4. What are **Association Rules**? Explain with an example.
5. List any **four applications of Unsupervised Learning**.

UNIT 9: Neural Network:

1. Define **Artificial Neural Network (ANN)**. Mention its advantages.
2. Differentiate between **Biological Neurons** and **Artificial Neurons**.
3. Explain any **three types of Activation Functions** used in Neural Networks.
4. What is **Back Propagation**? State its purpose.
5. What is **Deep Learning**? Mention any two applications.

SECTION B: 16 MARKS QUESTIONS (15 Questions)

UNIT 1: Introduction to Machine Learning:

1. Explain the **Overview of Human Learning and Machine Learning**. Compare both with suitable examples.
2. Describe the **Types of Machine Learning**:
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learningwith examples.
3. Explain the **Applications of Machine Learning** in different fields such as healthcare, finance, education, e-commerce, and transportation.
4. Discuss the **Tools and Technologies used in Machine Learning**, including their features and uses.
5. Compare **Human Learning and Machine Learning** based on:
 - Learning process
 - Decision-making
 - Adaptability
 - Speed
 - Accuracy

UNIT 2: Preparing to Model:

1. Explain the **Machine Learning process (activities)** with a neat diagram.
2. Describe the **Types and Structures of Data** used in Machine Learning with suitable examples.
3. Explain **Data Quality** and discuss different **Data Remediation techniques**.
4. Explain the **Data Preprocessing** techniques used in Machine Learning, including:
 - Data Cleaning
 - Data Transformation
 - Data Integration
 - Data Reduction
5. Explain the following feature selection techniques with suitable examples:
 - Dimensionality Reduction
 - Feature Subset Selection

UNIT 3: Modeling and Evaluation:

1. Explain the **process of selecting a Machine Learning model**. Differentiate between **Predictive** and **Descriptive** models with suitable examples.

2. Describe the **steps involved in training a supervised learning model** with a neat diagram.
3. Explain **Model Representation** and **Model Interpretability** with suitable examples.
4. Discuss the **different methods used to evaluate the performance of a Machine Learning model**.
5. Explain the **techniques used to improve the performance of a Machine Learning model**, such as:
 - Feature Selection
 - Hyperparameter Tuning
 - Cross-Validation
 - Handling Overfitting and Underfitting

UNIT 4: Basics of Feature Engineering:

1. Explain **Feature Engineering** and discuss its importance in Machine Learning.
2. Describe the **Feature Transformation** techniques:
 - Feature Construction
 - Feature Extractionwith suitable examples.
3. Explain **Feature Subset Selection** and discuss its role in improving Machine Learning models.
4. Discuss the **issues associated with high-dimensional data** and explain how they can be addressed.
5. Explain the **overall process of Feature Engineering**, including:
 - Key Drivers
 - Feature Measurement
 - Feature Selection
 - Feature Transformation

UNIT 5: Overview of Probability:

1. Explain the **Concepts of Probability** and discuss their importance in Machine Learning.
2. Describe **Random Variables** and explain the difference between **Discrete** and **Continuous Probability Distributions** with suitable examples.
3. Explain the **Central Limit Theorem (CLT)** and discuss its significance in statistical analysis and Machine Learning.
4. Explain the process of **Hypothesis Testing**, including:
 - Null Hypothesis (H_0)
 - Alternative Hypothesis (H_1)
 - Significance Level
 - Decision Making
5. Discuss the following statistical concepts used in Machine Learning:
 - Sampling Distributions
 - Multiple Random Variables
 - Monte Carlo Approximation

UNIT 6: Bayesian Concept Learning:

1. Explain the **importance of Bayesian methods** in Machine Learning with suitable examples.
2. State and explain **Bayes' Theorem**. Illustrate it with a numerical example.
3. Explain **Bayes' Theorem in Concept Learning** with a suitable example.
4. Describe the **Bayesian Belief Network (BBN)**, its components, working, advantages, and applications.
5. Compare **Bayesian Concept Learning** with traditional concept learning approaches.

UNIT 7: Supervised Learning:

1. Explain **Supervised Learning** with a neat diagram and suitable examples.
2. Describe the **Classification Model**, its learning steps, and applications.
3. Explain the **Classification Algorithms** used in Machine Learning, such as:
 - Decision Tree
 - Naïve Bayes
 - k-Nearest Neighbors (KNN)
 - Support Vector Machine (SVM)
4. Explain **Regression** and discuss different **Regression Algorithms**, such as:
 - Linear Regression
 - Polynomial Regression
 - Logistic Regression
 - Decision Tree Regression
5. Compare **Classification** and **Regression** based on:
 - Output
 - Data Type
 - Objective
 - Algorithms
 - Applications

UNIT 8: Unsupervised Learning:

1. Explain **Unsupervised Learning** with a neat diagram and suitable examples.
2. Compare **Supervised Learning** and **Unsupervised Learning** based on:
 - Training Data
 - Output
 - Goal
 - Algorithms
 - Applications
3. Explain **Clustering** in detail, including its working, advantages, and applications.

4. Describe **Association Rules** and explain their importance in Machine Learning with suitable examples.
5. Discuss the **Applications of Unsupervised Learning** in different domains such as:
 - Customer Segmentation
 - Market Basket Analysis
 - Recommendation Systems
 - Anomaly Detection
 - Image Segmentation

UNIT 9: Neural Network:

1. Explain the **Introduction to Neural Networks** and compare **Biological Neurons** with **Artificial Neurons** using a suitable diagram.
2. Describe the **Types of Activation Functions**:
 - Sigmoid
 - ReLU
 - Hyperbolic Tangent (Tanh)with their advantages and disadvantages.
3. Explain the **Architecture of an Artificial Neural Network (ANN)**, including:
 - Input Layer
 - Hidden Layer
 - Output Layerwith a neat diagram.
4. Explain the **Learning Process in ANN** and describe the **Back Propagation Algorithm** with a suitable diagram.
5. What is **Deep Learning**? Explain its architecture, advantages, and real-world applications.

SECTION C: MULTIPLE CHOICE QUESTIONS (30 Questions)

UNIT 1: Introduction to Machine Learning:

1. Machine Learning is a subset of:

- A) Data Mining
- B) Artificial Intelligence
- C) Cloud Computing
- D) Cyber Security

✓ **Answer: B**

2. Which type of Machine Learning uses labeled data for training?

- A) Unsupervised Learning
- B) Reinforcement Learning
- C) Supervised Learning
- D) Deep Learning

✓ **Answer: C**

3. Which of the following is a common application of Machine Learning?

- A) Spam Detection
- B) Weather Forecasting
- C) Recommendation Systems
- D) All of the Above

✓ **Answer: D**

4. Which of the following is a popular programming language used for Machine Learning?

- A) Python
- B) HTML
- C) CSS
- D) SQL

✓ **Answer: A**

5. Which type of Machine Learning learns by interacting with an environment using rewards and penalties?

- A) Supervised Learning
- B) Unsupervised Learning
- C) Reinforcement Learning
- D) Semi-Supervised Learning

✓ **Answer: C**

UNIT 2: Preparing to Model:

1. Which of the following is the first step in a Machine Learning project?

- A) Model Deployment
- B) Data Collection
- C) Model Evaluation
- D) Prediction

✓ **Answer: B**

2. Which data quality issue occurs when some values are unavailable?

- A) Duplicate Data
- B) Missing Data
- C) Noise
- D) Redundant Data

✓ **Answer: B**

3. Dimensionality Reduction is mainly used to:

- A) Increase the number of features
- B) Reduce the number of features while preserving important information
- C) Increase training time
- D) Create duplicate features

✓ **Answer: B**

4. Feature Subset Selection is used to:

- A) Remove irrelevant and redundant features
- B) Increase data size
- C) Create new datasets
- D) Label the data automatically

✓ **Answer: A**

5. Which of the following is an important step in Data Preprocessing?

- A) Data Cleaning
- B) Feature Selection
- C) Data Transformation
- D) All of the Above

✓ **Answer: D**

UNIT 3: Modeling and Evaluation:

1. Which type of model is mainly used to predict future outcomes?

- A) Descriptive Model
- B) Predictive Model
- C) Clustering Model
- D) Association Model

✓ **Answer: B**

2. In supervised learning, a model is trained using:

- A) Unlabeled Data
- B) Random Data
- C) Labeled Data
- D) Incomplete Data

✓ **Answer: C**

3. Model Interpretability refers to:

- A) Increasing the size of the dataset
- B) Understanding how a model makes predictions
- C) Reducing the number of features
- D) Improving hardware performance

✓ **Answer: B**

4. Which of the following is used to evaluate the performance of a Machine Learning model?

- A) Accuracy
- B) Precision
- C) Recall
- D) All of the Above

✓ **Answer: D**

5. Which technique is commonly used to improve the performance of a Machine Learning model?

- A) Feature Selection
- B) Hyperparameter Tuning
- C) Cross-Validation
- D) All of the Above

✓ **Answer: D**

UNIT 4: Basics of Feature Engineering:

1. Feature Engineering is the process of:

- A) Collecting data only
- B) Creating and selecting useful features for Machine Learning models
- C) Deploying Machine Learning models
- D) Storing data in databases

✓ **Answer: B**

2. Which of the following is a type of Feature Transformation?

- A) Feature Construction
- B) Feature Extraction
- C) Both A and B
- D) Model Evaluation

✓ **Answer: C**

3. High-dimensional data generally contains:

- A) Very few features
- B) A large number of features
- C) No missing values
- D) Only numerical data

✓ **Answer: B**

4. The main purpose of Feature Subset Selection is to:

- A) Increase the number of features
- B) Remove irrelevant and redundant features
- C) Increase data size
- D) Duplicate the dataset

✓ **Answer: B**

5. Which of the following is a benefit of Feature Engineering?

- A) Improves model accuracy
- B) Reduces training time
- C) Enhances model performance
- D) All of the Above

✓ **Answer: D**

UNIT 5: Overview of Probability:

Which branch of mathematics deals with uncertainty and chance?

- A) Algebra
- B) Probability
- C) Geometry
- D) Calculus

✓ **Answer: B**

2. A random variable that can take only finite or countable values is called:

- A) Continuous Random Variable
- B) Discrete Random Variable
- C) Normal Variable
- D) Independent Variable

✓ **Answer: B**

3. The Central Limit Theorem (CLT) states that the sampling distribution of the sample mean tends to be:

- A) Uniform Distribution
- B) Exponential Distribution
- C) Normal Distribution
- D) Poisson Distribution

✓ **Answer: C**

4. In hypothesis testing, the Null Hypothesis is represented by:

- A) H_0
- B) H_1
- C) H_2
- D) H_3

☒ **Answer: A**

5. Monte Carlo Approximation is mainly based on:

- A) Deterministic Calculations
- B) Random Sampling
- C) Database Queries
- D) Linear Regression

☒ **Answer: B**

UNIT 6: Bayesian Concept Learning:

1. Bayes' Theorem is mainly used to:

- A) Sort data
- B) Update probabilities based on new evidence
- C) Compress data
- D) Encrypt information

☒ **Answer: B**

2. Which probability represents the prior belief before observing evidence?

- A) Posterior Probability
- B) Conditional Probability
- C) Prior Probability
- D) Joint Probability

☒ **Answer: C**

3. A Bayesian Belief Network (BBN) is represented using:

- A) Tree Structure
- B) Directed Acyclic Graph (DAG)
- C) Linked List
- D) Binary Tree

☒ **Answer: B**

4. Bayes' Theorem is widely used in:

- A) Spam Detection
- B) Medical Diagnosis
- C) Email Filtering
- D) All of the Above

☒ **Answer: D**

5. Which of the following is an advantage of Bayesian methods?

- A) They can update predictions when new evidence is available.
- B) They always require large datasets.
- C) They cannot handle uncertainty.
- D) They work only for supervised learning.

✓ **Answer: A**

UNIT 7: Supervised Learning:

1. In Supervised Learning, the model is trained using:

- A) Unlabeled Data
- B) Labeled Data
- C) Random Data
- D) No Data

✓ **Answer: B**

2. Which Supervised Learning task predicts categorical outputs?

- A) Regression
- B) Classification
- C) Clustering
- D) Association

✓ **Answer: B**

3. Which algorithm is commonly used for Regression problems?

- A) K-Means Clustering
- B) Linear Regression
- C) Apriori Algorithm
- D) DBSCAN

✓ **Answer: B**

4. Which of the following is a Classification Algorithm?

- A) Decision Tree
- B) Naïve Bayes
- C) Support Vector Machine (SVM)
- D) All of the Above

✓ **Answer: D**

5. Predicting the price of a house is an example of:

- A) Classification
- B) Regression
- C) Clustering
- D) Reinforcement Learning

✓ **Answer: B**

UNIT 8: Unsupervised Learning:

1. Unsupervised Learning uses:

- A) Labeled Data
- B) Unlabeled Data
- C) Structured Data Only
- D) Test Data Only

✓ Answer: B

2. Which technique is mainly used to group similar data points?

- A) Regression
- B) Classification
- C) Clustering
- D) Prediction

✓ Answer: C

3. Market Basket Analysis is an application of:

- A) Regression
- B) Association Rules
- C) Classification
- D) Reinforcement Learning

✓ Answer: B

4. Which of the following is an application of Unsupervised Learning?

- A) Customer Segmentation
- B) Fraud Detection
- C) Market Basket Analysis
- D) All of the Above

✓ Answer: D

5. The main difference between Supervised and Unsupervised Learning is:

- A) Supervised Learning uses labeled data, while Unsupervised Learning uses unlabeled data.
- B) Both use labeled data.
- C) Unsupervised Learning requires output labels.
- D) Supervised Learning does not require training data.

✓ Answer: A

UNIT 9: Neural Network:

1. An Artificial Neural Network (ANN) is inspired by:

- A) Computer Networks
- B) Human Brain
- C) Operating System
- D) Database System

✓ Answer: B

2. Which activation function outputs values between 0 and 1?

- A) ReLU
- B) Sigmoid
- C) Tanh
- D) Linear

✓ Answer: B

3. Which algorithm is commonly used to train an Artificial Neural Network?

- A) K-Means
- B) Back Propagation
- C) Apriori
- D) Naïve Bayes

✓ Answer: B

4. Deep Learning mainly uses:

- A) Single-Layer Networks
- B) Multiple Hidden Layers
- C) Decision Trees
- D) Linear Regression

✓ Answer: B